

Electromagnetic spectrum and space

Teacher guide

Use color and reflection to introduce the electromagnetic spectrum. Connect everyday observations to the signals that radio telescopes detect.

Learning objectives

- Explain how reflected and absorbed light affect an object's appearance.
- Describe electromagnetic radiation and the electromagnetic spectrum.
- Explain why astronomers use radio telescopes.

Materials

- Visible-spectrum chart and colored stickers
- Electromagnetic-spectrum worksheet and movable term cards
- Colored pencils or crayons: red, blue, yellow, green, and purple

1 Explore the colors we see

1. Ask students how light reaches their eyes from a colored object. Discuss reflection and absorption.
2. Have students mix red and yellow pigments, then yellow and blue pigments. Compare the resulting colors.
3. Explain that mixing pigments differs from mixing colored light. Pigments change which wavelengths a surface reflects or absorbs.

2 Match reflection to the spectrum

Students work in pairs with a visible-spectrum chart. A timed challenge is optional; ask students to explain each placement.

1. Name the reflected colors in a sample. Students place stickers on those parts of the chart.
2. Name a color that the sample absorbs. Students identify that region and distinguish it from reflected light.
3. Discuss a surface that reflects most visible wavelengths, then one that absorbs most visible wavelengths.

Keep the light source the same when comparing samples. The light reaching our eyes depends on both illumination and the surface.

Connect light to astronomy

3 Discuss how we study space

Ask what students already know about telescopes. Explain that visible-light telescopes collect light our eyes can detect.

Radio telescopes collect another part of the electromagnetic spectrum. Different wavelength ranges reveal different properties of astronomical objects.



Figure 1. The Sun provides a familiar source of light and radiant energy.

4 Organize the electromagnetic spectrum

1. Introduce electromagnetic radiation as energy carried by electromagnetic waves. Connect the term to electric and magnetic fields.
2. Have students arrange the term cards on their spectrum worksheet. Locate the visible-light region and color it.
3. Locate radio waves. Discuss why our eyes cannot directly detect this radiation.

Discussion and reflection

- Which light reaches your eyes when you look at a colored object?
- Where do visible light and radio waves fit in the electromagnetic spectrum?
- What could a radio telescope reveal that a visible-light telescope does not show?

Sources

Adapted from the DSPIRA electromagnetic spectrum and space teaching outline. Original Sun illustration retained.

[NASA: Wave behaviors](#)

[NASA JPL: The science of color](#)